(Please write your Enrolment No. immediately) Enrolment No.....

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

3RD Semester, November 2022

Paper Code:ECC-207 Subject: Digital Logic and Computer Design

Time: 1.5 hrs. Max Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Question 1

a. How is BCD Code different from Binary Code? (2 Marks) CO1
b. Explain addition of two negative numbers in 2's compliment form with help of an example. (2 Marks) CO1
c. Implement two input EX-OR gate using a Multiplexer? (2 Marks) CO1
d. Explain the concept of Level Triggering in Sequential Circuits? (2 Marks) CO2
e. How is state diagram different from state table? (2 Marks) CO 2

Question 2

a. Differentiate between Serial Adder and Binary Parallel Adder? (3 Marks), CO 1,2
b. Explain how to Convert SR Flip Flop into D Flip Flop. (3 Marks), CO 2
c. Design a 4 bit Ripple Down Counter and draw the waveform? (4 Marks), CO 2

Question 3

a. For the Boolean expression $F = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC + AB\bar{C}$ find out the minimized Product of Sum (POS) expression (use K-map) (5 Marks), CO1
b. What is a Multiplexer? Design a 16:1 multiplexer using 4:1 multiplexers? (5 Marks), CO1

c. **Question 4**

a. Using K-map simplify expression, Y= f(A,B,C,D) = $\pi(0,1,2,4,5,8,9,15)$ (2 Marks), CO 1
b. A combinational circuit is defined by function $F_1 = \sum_m (1, 5, 7)$, $F_2 = \sum_m (5, 6, 7)$. Implement the circuit with a PLA. (3 Marks), CO 2
c. Design a Mod-5 asynchronous counter using T-FF. (5 Marks), CO 2